

Coherent perfect absorption of surface plasmon resonances

Thin film

We derive the equations in Ultrathin broadband nearly perfect absorber with symmetrical coherent illumination (<https://opg.optica.org/abstract.cfm?uri=oe-20-3-2246>) by **Pu et al.**

We take a slab of index n , thickness d , finite in the x direction and centered at $x = 0$. We take the electric field to be polarized along the y direction,

$$\mathbf{E}(\mathbf{r}, \omega) = (Ae^{ik(x+d/2)} + De^{-ik(x+d/2)})\mathbf{e}_y \text{ for } x < -d/2,$$

$$\mathbf{E}(\mathbf{r}, \omega) = (Ce^{ik(x-d/2)} + Be^{-ik(x-d/2)})\mathbf{e}_y \text{ for } x > d/2 \text{ and}$$

$$\mathbf{E}(\mathbf{r}, \omega) = (E_1e^{ipx} + E_2e^{-ipx})\mathbf{e}_y \text{ for } |x| < d/2.$$

By the wave equation, we have that the wavevectors are given by $k(\omega) = \omega/c$ and $p(\omega) = n\omega/c$.

The electric field cannot change across the boundaries (since it has no normal component), which means that

$$A + D = E_1e^{-ipd/2} + E_2e^{ipd/2} \text{ and } C + B = E_1e^{ipd/2} + E_2e^{-ipd/2}$$

Our next boundary conditions come from the magnetic field, $\mathbf{H}(\mathbf{r}, \omega)$.

We find $\mathbf{H}(\mathbf{r}, \omega)$ by noting that $\mathbf{H}(\mathbf{r}, \omega) = \frac{1}{i\omega\mu_0} \nabla \times \mathbf{E}(\mathbf{r}, \omega)$, and, because the electric field is purely polarized along the $\mathbf{e}_y$ direction and only has an x dependence, we have $\mathbf{H}(\mathbf{r}, \omega) = (-i/\omega\mu_0)\partial_x(\mathbf{e}_y \cdot \mathbf{E}(\mathbf{r}, \omega))\mathbf{e}_z$. Writing this for the different regions, we have:

$$\mathbf{H}(\mathbf{r}, \omega) = \frac{k}{\omega\mu_0}(Ae^{ik(x+d/2)} - De^{-ik(x+d/2)})\mathbf{e}_z \text{ for } x < d/2,$$

$$\mathbf{H}(\mathbf{r}, \omega) = \frac{k}{\omega\mu_0}(Ce^{ik(x-d/2)} - Be^{-ik(x-d/2)})\mathbf{e}_z \text{ for } x > d/2 \text{ and}$$

$$\mathbf{H}(\mathbf{r}, \omega) = \frac{p}{\omega\mu_0}(E_1e^{ipx} - E_2e^{-ipx})\mathbf{e}_z \text{ for } |x| < d/2.$$

Thus, requiring the magnetic field's transverse component to remain unchanged across the boundaries (since there is no surface current), gives us:

$$k(A - D) = p(E_1 e^{-ipd/2} - E_2 e^{ipd/2}) \rightarrow A - D = \frac{p}{k}(E_1 e^{-ipd/2} - E_2 e^{ipd/2}) \text{ and}$$

$$k(C - B) = p(E_1 e^{ipd/2} - E_2 e^{-ipd/2}) \rightarrow C - B = \frac{p}{k}(E_1 e^{ipd/2} - E_2 e^{-ipd/2}).$$

Therefore, we have the following system of four equations:

$$C = (1/2)[E_1 e^{ipd/2}(1 + p/k) + E_2 e^{-ipd/2}(1 - p/k)]$$

$$B = (1/2)[E_1 e^{ipd/2}(1 - p/k) + E_2 e^{-ipd/2}(1 + p/k)]$$

$$A = (1/2)[E_1 e^{-ipd/2}(1 + p/k) + E_2 e^{ipd/2}(1 - p/k)]$$

$$D = (1/2)[E_1 e^{-ipd/2}(1 - p/k) + E_2 e^{ipd/2}(1 + p/k)]$$

We note that this system of equations may be written as two separate matrix equations relating the amplitudes, A, D to the amplitudes E_1, E_2 (the first matrix) and relating the amplitudes, B, C to the amplitudes E_1, E_2 . Thus, we may herefore, we may write the following matrix equation relating the amplitudes A, D with the amplitudes C, B :

$$\begin{bmatrix} C \\ B \end{bmatrix} = \begin{bmatrix} e^{ipd/2}(1 + p/k) & e^{-ipd/2}(1 - p/k) \\ e^{ipd/2}(1 - p/k) & e^{-ipd/2}(1 + p/k) \end{bmatrix} \begin{bmatrix} e^{-ipd/2}(1 + p/k) & e^{ipd/2}(1 - p/k) \\ e^{-ipd/2}(1 - p/k) & e^{ipd/2}(1 + p/k) \end{bmatrix}^{-1} \times \begin{bmatrix} A \\ D \end{bmatrix}$$

We write this matrix product as:

$$\frac{k}{4p} \begin{bmatrix} e^{ipd/2}(1 + p/k) & e^{-ipd/2}(1 - p/k) \\ e^{ipd/2}(1 - p/k) & e^{-ipd/2}(1 + p/k) \end{bmatrix} \begin{bmatrix} e^{ipd/2}(1 + p/k) & -e^{ipd/2}(1 - p/k) \\ -e^{-ipd/2}(1 - p/k) & e^{-ipd/2}(1 + p/k) \end{bmatrix}$$

Note also that from our previous definitions of the wavevectors, we have $p = nk$, which allows us to simplify:

$$C = \frac{1}{4n} \left[(e^{inkd}(1 + n)^2 - e^{-inkd}(1 - n)^2)A + (e^{-inkd} - e^{inkd})(1 - n^2)D \right]$$

$$B = \frac{1}{4n} \left[(e^{inkd} - e^{-inkd})(1 - n^2)A + (e^{-inkd}(1 + n)^2 - e^{inkd}(1 - n)^2)D \right]$$

We us the second equation to solve for D in terms of the amplitudes A, B :

$$D = \frac{4nB - (1 - n^2)(e^{inkd} - e^{-inkd})A}{e^{-inkd}(1 + n)^2 - e^{inkd}(1 - n)^2}$$

So the reflection coefficient (relating the amplitude, D to the amplitude, A) for normal incidence is given by:

$$r(\omega) = \frac{(1 - n^2)(e^{-inkd} - e^{inkd})}{(1 + n)^2 e^{-inkd} - (1 - n)^2 e^{inkd}} = \frac{(n^2 - 1)(e^{2inkd} - 1)}{(1 + n)^2 - (n - 1)^2 e^{2inkd}},$$

which is in agreement with **Equation 2** of the paper by **Pu et al.** And we get that the transmission coefficient (relating the amplitude C to the amplitude A) is given by:

$$t(\omega) = \frac{e^{inkd}(1 + n)^2 - e^{-inkd}(1 - n)^2}{4n} + \frac{(1 - n^2)^2(e^{inkd} - e^{-inkd})^2}{4n(e^{-inkd}(1 + n)^2 - e^{inkd}(1 - n)^2)}$$

We put these under a common denominator to obtain:

$$\frac{1}{4n} \left[\frac{(1 + n)^4 + (1 - n)^4 - (1 - n^2)^2(e^{2inkd} + e^{-2inkd}) + (1 - n^2)^2(e^{2inkd} - 2 + e^{-2inkd})}{e^{-inkd}(1 + n)^2 - e^{inkd}(1 - n)^2} \right].$$

This simplifies to (we note that in the combination $(1 + n)^4 + (1 - n)^4$ we do not have any non-zero contribution from odd powers of n):

$$t(\omega) = \frac{1}{4n} \left[\frac{2 + 12n^2 + 2n^4 - 2(1 - 2n^2 + n^4)}{e^{-inkd}(1 + n)^2 - e^{inkd}(1 - n)^2} \right] = \frac{4ne^{inkd}}{(n + 1)^2 - (n - 1)^2 e^{2inkd}},$$

which is consistent with **Equation 3** of **Pu et al.**

Sanity check on the reflection/transmission coefficients

$|r(\omega)|^2 + |t(\omega)|^2$ should equal unity if there is no dissipation in the slab (i.e. if n is entirely real valued).

$$\begin{aligned} |r(\omega)|^2 + |t(\omega)|^2 &= \frac{16n^2 + (n^2 - 1)^2(e^{2inkd} - 1)(e^{-2inkd} - 1)}{|(n + 1)^2 - (n - 1)^2 e^{2inkd}|^2} = \\ &= \frac{16n^2 + (n^2 - 1)^2(2 - e^{2inkd} - e^{-2inkd})}{(n + 1)^4 + (n - 1)^4 - (n^2 - 1)^2(e^{2inkd} + e^{-2inkd})} \end{aligned}$$

The denominator can be rewritten as: $2n^4 + 12n^2 + 2 - (n^2 - 1)^2(e^{2inkd} + e^{-2inkd})$

The numerator can be rewritten as $2n^4 - 4n^2 + 2 + 16n^2 - (n^2 - 1)^2(e^{2inkd} + e^{-2inkd})$

Therefore, we clearly have $|r(\omega)|^2 + |t(\omega)|^2 = 1$ in the absence of materials losses.

Similarly, we must have $|r(\omega)|^2 + |t(\omega)|^2 = 1$ if n is purely imaginary (corresponding to a completely real dielectric constant but with $\varepsilon < 0$). In this case, we have (since $n^* = -n$):

$$\begin{aligned} |r(\omega)|^2 + |t(\omega)|^2 &= \frac{-16n^2e^{2inkd} + (n^2 - 1)^2(e^{2inkd} - 1)^2}{\left[(1 + n)^2 - (n - 1)^2e^{2inkd}\right] \left[(1 - n)^2 - (n + 1)^2e^{2inkd}\right]} = \\ &= \frac{-16n^2e^{2inkd} + (n^2 - 1)^2(e^{4inkd} + 1 - 2e^{2inkd})}{(n^2 - 1)^2 \left(1 + e^{4inkd}\right) - [(n + 1)^4 + (n - 1)^4]e^{2inkd}} = \\ &= \frac{-2(8n^2 + (n^2 - 1)^2)e^{2inkd} + (n^2 - 1)^2(e^{4inkd} + 1)}{(n^2 - 1)^2 \left(1 + e^{4inkd}\right) - 2[n^4 + 6n^2 + 1]e^{2inkd}} = \\ &= \frac{-2[n^4 + 6n^2 + 2]e^{2inkd} + (n^2 - 1)^2(e^{4inkd} + 1)}{(n^2 - 1)^2 \left(1 + e^{4inkd}\right) - 2[n^4 + 6n^2 + 1]e^{2inkd}} = 1, \end{aligned}$$

as required.

Coherent perfect absorption for a thin film

We must have a zero in the determinant of the S-matrix:

$$t^2 - r^2 = 0 \rightarrow (t - r)(t + r) = 0$$

For the case, $t + r$,

$$\begin{aligned} \frac{(n^2 - 1)(e^{2inkd} - 1)}{(1 + n)^2 - (n - 1)^2e^{2inkd}} + \frac{4ne^{inkd}}{(n + 1)^2 - (n - 1)^2e^{2inkd}} &= 0 \rightarrow \\ (n^2 - 1)(e^{2inkd} - 1) + 4ne^{inkd} &= 0 \end{aligned}$$

For brevity, we set $y = e^{inkd}$ to obtain:

$$(y^2 - 1)(n^2 - 1) + 4ny = 0 \rightarrow y = \frac{-4n \pm \sqrt{16n^2 + 4(n^2 - 1)^2}}{2(n^2 - 1)} =$$

$$\frac{-2n \pm (n^2 + 1)}{(n^2 - 1)} = \frac{n - 1}{n + 1}, -\frac{n + 1}{n - 1}$$

However, note that the second root is not possible, since $(n + 1)/(n - 1)$ will have a magnitude greater than one and if $n'' > 0$, we will necessarily have $|y| < 1$.

Coherent perfect absorption for a metallic cylinder

We consider TM modes ($\mathbf{H} = H\mathbf{e}_z$), defined in [Perfect coupling of light to surface plasmons by coherent absorption](https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.108.186805)

(<https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.108.186805>), as modes with electric field in the x-y plane. We solve for the field profile inside and outside the cylinder.

Maxwell's equations read:

$$\nabla \times \mathbf{H}(\mathbf{r}, \omega) = -i\omega\epsilon_0\epsilon(\mathbf{r})\mathbf{E}(\mathbf{r}, \omega), \nabla \times \mathbf{E}(\mathbf{r}, \omega) = i\omega\mu_0\mathbf{H}(\mathbf{r}, \omega).$$

We first focus on the region outside of the cylinder, $\epsilon(\mathbf{r}) = 1$. Here, we have that $\nabla \times \nabla \times \mathbf{H}(\mathbf{r}, \omega) = (\omega^2/c^2)\mathbf{H} \rightarrow \nabla^2\mathbf{H}(\mathbf{r}, \omega) = -(\omega^2/c^2)\mathbf{H}(\mathbf{r}, \omega)$. We write this equation in cylindrical coordinates:

$$\frac{1}{\rho}\partial_\rho(\rho\partial_\rho H(\rho, \phi)) + \frac{1}{\rho^2}\partial_\phi^2 H(\rho, \phi) = -(\omega^2/c^2)H(\rho, \phi)$$

We take $H(\rho, \phi) = e^{im\phi}H(\rho)$, which gives us:

$$\frac{1}{\rho}(\partial_\rho H(\rho) + \rho\partial_\rho^2 H(\rho)) = (m^2/\rho^2 - \omega^2/c^2)H(\rho)$$

Equivalently,

$$\rho^2\partial_\rho^2 H(\rho) + \rho\partial_\rho H(\rho) + (\rho^2\omega^2/c^2 - m^2)H(\rho) = 0$$

We define $H(\rho) = y(\rho\omega/c)$ which gives us:

$$(\rho\omega/c)^2 y''(\rho\omega/c) + (\rho\omega/c)y'(\rho\omega/c) + (\rho^2\omega^2/c^2 - m^2)y(\rho\omega/c) = 0$$

This is the (unmodified) Bessel equation. Therefore, the solutions may be written either in terms of the Bessel functions of the first or second kind or in terms of the Hankel functions, which are simply functions whose real and imaginary parts correspond to the Bessel functions of the first and second kind, respectively, $H^{(1,2)} = J \pm iY$, corresponding to incoming waves for $H^{(2)}$ and outgoing (scattered) waves for $H^{(1)}$.

The field inside the cylinder can only be written in terms of the Bessel function of the first kind, since the field must be finite at the origin. Therefore, we have:

$$\mathbf{H}(\mathbf{r}, \omega) \cdot \mathbf{e}_z = e^{im\phi} (H_m^{(2)}(\rho\omega/c) + sH_m^{(1)}(\rho\omega/c)) \text{ for } \rho > R$$

and

$$\mathbf{H}(\mathbf{r}, \omega) \cdot \mathbf{e}_z = ae^{im\phi} J_m(n\rho\omega/c) \text{ for } \rho < R.$$

Therefore, we have that $aJ_m(Rn\omega/c) = H_m^{(2)}(R\omega/c) + sH_m^{(1)}(R\omega/c)$

To find additional boundary conditions, we need to find the electric field,

$$\mathbf{E}(\mathbf{r}, \omega) = \frac{i}{\omega\epsilon_0\epsilon(\mathbf{r})} \nabla \times \mathbf{H}(\mathbf{r}, \omega)$$

where the curl in cylindrical coordinates may be written as:

$$\nabla \times \mathbf{H} = \frac{1}{\rho} \partial_\phi (H(\rho, \phi)) \mathbf{e}_\rho - \partial_\rho (H(\rho, \phi)) \mathbf{e}_\phi$$

We require that the transverse component ($\mathbf{e}_\phi$ component) of the electric field remain unchanged across the boundary. In addition, we require that the normal component of the displacement field not change. Let's first examine the second condition. This essentially is the same as the magnetic field boundary condition, since we get that $(im/\rho)H(\rho, \phi)$ must match. Therefore, the only nontrivial boundary condition comes from the transverse component of the electric field:

$$\left[(H_m^{(2)}(R\omega/c))' + s(H_m^{(1)}(R\omega/c))' \right] = \frac{an}{\epsilon} (J_m(nR\omega/c))' = \frac{a}{n} (J_m(nR\omega/c))'$$

Combining the two boundary conditions then gives us:

$$\frac{nJ_m(nR\omega/c)}{(J_m)'(nR\omega/c)} \left[(H_m^{(2)}(R\omega/c))' + s(H_m^{(1)}(R\omega/c))' \right] = H_m^{(2)}(R\omega/c) + sH_m^{(1)}(R\omega/c)$$

Therefore, by isolating all terms that multiply the scattering amplitude, s , we get the following:

$$nJ_m(nR\omega/c)(H_m^{(2)}(R\omega/c))' - (J_m(nR\omega/c))'H_m^{(2)}(R\omega/c) = s \left[H_m^{(1)}(R\omega/c)(J_m(nR\omega/c))' - nJ_m(nR\omega/c)(H_m^{(1)}(R\omega/c))' \right]$$

And we obtain an expression for s ,

$$s = \frac{nJ_m(nR\omega/c)(H_m^{(2)}(R\omega/c))' - (J_m(nR\omega/c))'H_m^{(2)}(R\omega/c)}{H_m^{(1)}(R\omega/c)(J_m(nR\omega/c))' - nJ_m(nR\omega/c)(H_m^{(1)}(R\omega/c))'}$$

which is **Equation 2** in the paper by **Noh et al** which was cited at the beginning of this section.

Impossible coherent perfect absoropton for TE modes of a cylinder

For TE modes, we take $\mathbf{E}(\mathbf{r}, \omega) = E(\rho, \phi)\mathbf{e}_z$. One boundary condition equation, is, again,

$$aJ_m(Rn\omega/c) = H_m^{(2)}(R\omega/c) + sH_m^{(1)}(R\omega/c),$$

But this time this corresponds to boundary conditions on the electric field.

The magnetic field is found from:

$$-\frac{i}{\omega\mu_0} \nabla \times \mathbf{E}(\mathbf{r}, \omega) = \mathbf{H}(\mathbf{r}, \omega)$$

Using our previous expression for the curl of a vector function that has only an $\mathbf{e}_z$ dependence, reiterated below,

$$\nabla \times \mathbf{E}(\mathbf{r}, \omega) = \frac{1}{\rho} \partial_\phi(E(\rho, \phi))\mathbf{e}_\rho - \partial_\rho(E(\rho, \phi))\mathbf{e}_\phi,$$

and focusing on the $\mathbf{e}_\phi$ component of this relation, we obtain the following boundary condition:

$$an(J_m(nR\omega/c))' = (H_m^{(2)}(R\omega/c))' + s(H_m^{(1)}(R\omega/c))'$$

Combining the two boundary conditions then gives

$$\frac{J_m(nR\omega/c)}{n(J_m(nR\omega/c))'} \left[(H_m^{(2)}(R\omega/c))' + s(H_m^{(1)}(R\omega/c))' \right] = H_m^{(2)}(R\omega/c) + sH_m^{(1)}(R\omega/c)$$

From which we obtain the scattering coefficient:

$$s = \frac{nH_m^{(2)}(R\omega/c)J_m'(nR\omega/c) - H_m^{(2)}(R\omega/c)'J_m(nR\omega/c)}{H_m^{(1)}(R\omega/c)'J_m(nR\omega/c) - nH_m^{(1)}(R\omega/c)J_m'(nR\omega/c)}$$

Coherent perfect absorption for a sphere

We again consider TM modes. We approach this problem following **Jackson's Electrodynamics (page 429)**.

Consider the quantity $\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega)$. Since we have that

$$\nabla^2(\mathbf{r} \cdot \mathbf{E}) = \partial_i^2(r_j E_j) = (\partial_i^2 r_j)E_j + r_j \partial_i^2 E_j + 2(\partial_i r_j)(\partial_i E_j) = r_j \partial_i^2 E_j + 2\delta_{ij} \partial_i E_j = 2\nabla \cdot \mathbf{E} + \mathbf{r} \cdot \nabla^2 \mathbf{E}$$

We see that in a source-free region (i.e. where $\nabla \cdot \mathbf{E}(\mathbf{r}, \omega) = 0$) we must necessarily have by taking the dot product of $\mathbf{r}$ with the vector Helmholtz equation:

$$\mathbf{r} \cdot \nabla^2 \mathbf{E}(\mathbf{r}, \omega) + (\omega/c)^2 \mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega) = 0 \rightarrow \nabla^2(\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega)) + (\omega/c)^2(\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega)) = 0$$

Therefore, $\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega)$ satisfies the scalar Helmholtz equation. By similar manipulations, we also have that $\mathbf{r} \cdot \mathbf{H}(\mathbf{r}, \omega)$ satisfies the scalar Helmholtz equation.

Now we note that

$$\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega) = \frac{i}{\omega \epsilon_0 \epsilon} \mathbf{r} \cdot (\nabla \times \mathbf{H}(\mathbf{r}, \omega))$$

$$\text{We note that } \mathbf{r} \cdot (\nabla \times \mathbf{H}(\mathbf{r}, \omega)) = r_i \epsilon_{ijk} \partial_j H_k = (\mathbf{r} \times \nabla) \cdot \mathbf{H}(\mathbf{r}, \omega)$$

Therefore, we have that $\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega) = -(\mathbf{L} \cdot \mathbf{H}(\mathbf{r}, \omega))/(\varepsilon_0 \varepsilon \omega)$. From this, we see that the angular momentum operator acting on $\mathbf{H}(\mathbf{r}, \omega)$ must give something that satisfies the Helmholtz equation. From this, we find that $\mathbf{H}(\mathbf{r}, \omega) = \mathbf{L}Y_{lm}(\theta, \phi)$ multiplied by some spherical Bessel or Hankel function. Alternatively, we could have $\mathbf{E}(\mathbf{r}, \omega) = \mathbf{L}Y_{lm}(\theta, \phi)$. In the former case (TM modes), we have $\mathbf{r} \cdot \mathbf{H}(\mathbf{r}, \omega) = 0$ and in the latter case, we have $\mathbf{r} \cdot \mathbf{E}(\mathbf{r}, \omega) = 0$.

We consider TM modes. Within the sphere, we have $\mathbf{H}(\mathbf{r}, \omega) = aj_l(\varepsilon \omega r/c) \mathbf{L}Y_{lm}(\theta, \phi)$

and outside the sphere, we have:

$$\mathbf{H} = (sh_l^{(1)}(\omega r/c) + h_l^{(2)}(\omega r/c)) \mathbf{L}Y_{lm}(\theta, \phi)$$

$$\text{Therefore, we have } sh_l^{(1)}(\omega R/c) + h_l^{(2)}(\omega R/c) = aj_l(\sqrt{\varepsilon} \omega R/c)$$

and we have that (by continuity of the electric field's tangential component):

$$s \partial_r (r h_l^{(1)}(\omega R/c)) + \partial_r (r h_l^{(2)}(\omega R/c)) = a \partial_r (r j_l(\sqrt{\varepsilon} \omega R/c)) / \varepsilon$$

We look at CPA ($s = 0$) and use Stone and Noh's notation to write:

$$\partial_x (x h_l^{(2)}(x)) = a \partial_y (y j_l(y)) / \varepsilon$$

Putting everything together, we obtain:

$$\varepsilon \partial_x (x h_l^{(2)}(x)) j_l(y) = h_l^{(2)}(x) \partial_y (y j_l(y))$$

Appendix 1: Angular momentum

$$\mathbf{L} = ((L_+ + L_-)/2, (L_+ - L_-)/(2i), L_z)$$

$$\mathbf{e}_r = (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$$

Therefore, as can easily be verified, $\mathbf{e}_r \cdot \mathbf{L}$ is zero.

Appendix 2: Scalar Helmholtz equation for sphere

$$\text{We have } \nabla^2 \psi + (\omega/c)^2 \psi = 0$$

Separating variables, we obtain:

$$\frac{1}{r^2 f} \partial_r (r^2 \partial_r f) + \frac{1}{r^2 \sin(\theta) Y} \partial_\theta (\sin(\theta) \partial_\theta Y) + \frac{1}{r^2 \sin^2(\theta) Y} \partial_\phi^2 Y + (\omega/c)^2 = 0$$

This then gives us:

$$\frac{1}{f} \partial_r (r^2 \partial_r f) + ((\omega/c)^2 r^2 - l(l+1)) = 0$$

$$r^2 \partial_r^2 f + 2r \partial_r f + ((\omega/c)^2 r^2 - l(l+1)) f = 0$$

Therefore, we see that this is almost Bessel equation's with $x = (\omega/c)r$

we write $f = j/\sqrt{x}$, which gives us:

$$x^{3/2} \partial_x^2 j + x^{-1/2} (3/4) j - x^{1/2} \partial_x j + 2x^{1/2} \partial_x j - x^{-1/2} j + (x^2 - l(l+1)) j / \sqrt{x} = 0$$

Which is equivalent to:

$$x^2 \partial_x^2 j + x \partial_x j - j/4 + (x^2 - l(l+1)) j = 0$$

Equivalently,

$$x^2 \partial_x^2 j + x \partial_x j + (x^2 - (l+1/2)(l+1/2)) j = 0$$

Appendix 3: Static limit

In the static limit, we find:

$$r^2 \partial_r^2 f + 2r \partial_r f - l(l+1) f = 0$$

We take as an ansatz, $f = r^n$, which gives us:

$$n(n-1) + 2n - l(l+1) = 0 \rightarrow n(n+1) = l(l+1) \rightarrow n = l, -l-1$$